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## Quick Start Guide

Get up and running with Enphase Monitor in 5 minutes!

This guide will help you: 1. Set up your configuration file 2. Complete OAuth authentication 3. Configure your systems 4. Run your first query

### Step 1: Prerequisites

Make sure you have: - [ ] Go 1.21+ installed (go version to check) - [ ] Enphase Developer Portal account at <https://developer-v4.enphase.com/> - [ ] API credentials from the Developer Portal: - API Key - Client ID - Client Secret - [ ] System IDs for your Enphase installations (see Step 3)

### Step 2: Initial Setup

```
# Navigate to the project directory
cd enphase-monitor

# Create your configuration file
make setup
# OR manually:
# cp config.yaml.example config.yaml
```

### Step 3: Configure Your Systems

Edit config.yaml with your favorite text editor:

```
api:
  key: "YOUR_API_KEY"           # From Developer Portal
  client_id: "YOUR_CLIENT_ID"   # From Developer Portal
  client_secret: "YOUR_CLIENT_SECRET" # From Developer Portal
```

```
authorization_url: "https://api.enphaseenergy.com/oauth/token"
redirect_uri: "http://localhost:8080/callback"
refresh_token: "" # Will be filled in Step 4
```

systems:

```
- name: "Left Subpanel"      # Give it a friendly name
  id: "1234567"              # Your first system ID (see below)

- name: "Right Subpanel"     # Name your second system
  id: "7654321"              # Your second system ID
```

```
refresh_interval: 3600      # Query every hour (recommended)
```

**Finding Your System IDs:** See Finding Your System IDs in the README for detailed instructions.

## Step 4: Complete OAuth Setup

**This step is required before you can use the application!**

The application uses OAuth 2.0 for authentication. You need to complete a one-time OAuth setup:

```
./enphase-monitor --setup-oauth
```

This interactive wizard will: 1. Generate an authorization URL 2. Guide you to authorize the application in your browser 3. Help you exchange the authorization code for tokens 4. Show you the `refresh_token` to add to your config

**After the wizard completes:** 1. Copy the `refresh_token` from the output 2. Add it to your `config.yaml`: `api: refresh_token: "YOUR_REFRESH_TOKEN" # Paste here`

**Want to understand OAuth better?** See `OAUTH_SETUP.md` for a detailed explanation of how OAuth works, what each component does, and how authentication is performed.

## Step 5: Build and Install

```
# Install dependencies
```

```
go mod download
```

```
# Build the application
```

```
go build -o enphase-monitor
```

## Step 6: Test It!

Run a single query to make sure everything works:

```
./enphase-monitor --once
```

You should see output showing your combined system metrics!

## Step 7: Start Monitoring

Start continuous monitoring (refreshes every hour by default):

```
make run
```

Press **Ctrl+C** to stop.

## Common First-Time Issues

### “api configuration required”

→ Make sure you have filled in `api.key`, `api.client_id`, and `api.client_secret` in `config.yaml`

### “API request failed with status 401”

→ Your refresh token might be missing or expired. Complete OAuth setup: `./enphase-monitor --setup-oauth`

### “system must have id”

→ You need to replace the example system IDs with your actual ones from Enlighten (see Step 3)

### “redirect\_uri mismatch”

→ The redirect URI in your config must match exactly what you registered in the Enphase Developer Portal

## Next Steps

Once it is working:

- Query historical data: `./enphase-monitor --once --date 2026-01-15`
- Run on startup: Add to cron or systemd
- Build a dashboard: Parse the output or extend the code
- Set up alerts: Monitor grid dependence and trigger notifications

## Need More Help?

- **README.md** - Complete documentation with all features and options
- **OAUTH\_SETUP.md** - Detailed OAuth guide with troubleshooting
- **ARCHITECTURE.md** - Understand how the codebase works
- **Troubleshooting section** in **README.md**

Happy monitoring!